

Three tangent circles and a fair coin

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Abstract

Three circles touch in pairs, and a triangle is formed by choosing one point at random on each. The triangle contains the incentre of the triangle of centres with probability exactly $\frac{1}{2}$, whatever the three radii. The proof rests on one lemma about two touching circles: a random chord joining them crosses their common tangent at a point from which the two centres are seen at an angle that is uniformly distributed. A refinement turns the pair of random points into a uniform point on a sphere, on which each way of missing the incentre becomes a lune.

1 A ring of three circles

Draw three circles, each touching the other two, and pick a point at random on each of them. How often does the triangle of the three points contain the incentre of the triangle formed by the centres? It is worth guessing before reading on, and a few sketches like Figure 1 suggest that the answer is not obvious.

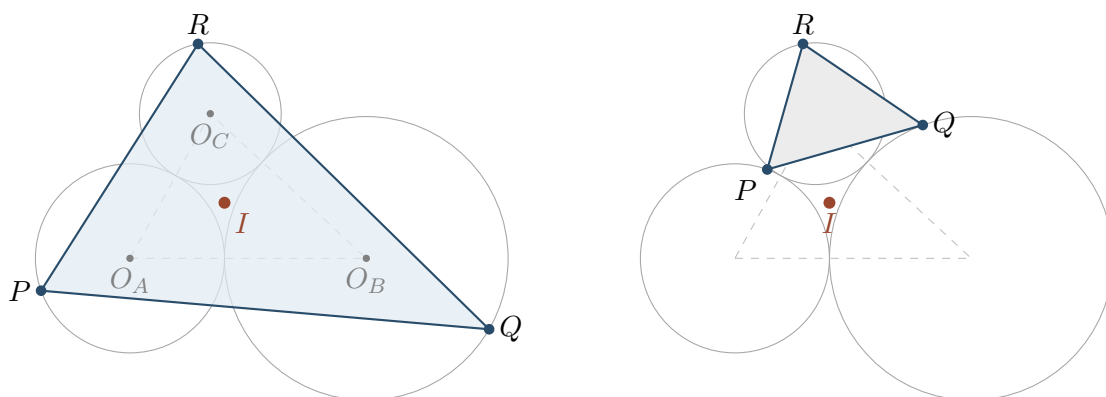


Figure 1: Radii 1, 1.5 and 0.75. On the left the random triangle contains the incentre I of the triangle of centres $O_A O_B O_C$; on the right it does not.

The answer is a fair coin.

Theorem 1. *For any three mutually tangent circles, with one uniform random point on each, chosen independently, the triangle of the three points contains the incentre of the triangle of centres with probability $\frac{1}{2}$.*

Dan asked this on Mathematics Stack Exchange for three equal circles [1], and there it was settled by an integral: Dan reduced it to a single integral, and user170231 evaluated it with a contour argument [2]. Dan then asked about unequal radii, where simulations gave $\frac{1}{2}$ again [3], and isolated a statement about two circles that would explain it [4]. On MathOverflow [5] he asked for a proof that shows why the answer is $\frac{1}{2}$. This note proves the theorem for all radii by proving that two-circle statement, in the following form.

Theorem 2. *Two circles touch at T , and a chord PQ joins a uniform random point P of one to an independent uniform random point Q of the other. Let Z be the point where PQ crosses the common tangent at T . Then Z lies on either side of T with probability $\frac{1}{2}$, and the angle $\theta = \angle O_A Z O_B$ at which Z sees the two centres is uniformly distributed on $(\frac{\pi}{2}, \pi)$.*

The angle is π when $Z = T$, and it falls as Z moves away; Theorem 2 says that it falls at a uniform rate in probability. Seen this way the radii drop out entirely, and that is why the final answer does not depend on them. Section 5 shows that θ has a hidden partner: together they describe a uniform point on a sphere.

2 What the incentre sees

Let the circles have centres O_A, O_B, O_C and radii a, b, c , and write A, B, C for the angles of the triangle of centres. Its sides are $b + c, c + a$ and $a + b$.

Lemma 1. *The incircle of $O_A O_B O_C$ touches its sides at the three points where the circles touch. Seen from its centre I , the three circles fill three angular sectors of widths $\pi - A, \pi - B, \pi - C$, which tile the full turn.*

Proof. The tangent from O_A to the incircle has length $\frac{1}{2}((a + b) + (a + c) - (b + c)) = a$, so the incircle touches $O_A O_B$ at the point T_C where the circles about O_A and O_B touch, and similarly at T_A and T_B . The radius IT_C is perpendicular to $O_A O_B$, so the line IT_C is the common tangent of those two circles; likewise IT_B is tangent to the circle about O_A . So from I , which lies outside every circle, the circle about O_A fills exactly the angle between the rays IT_C and IT_B . In the quadrilateral $O_A T_C I T_B$ two angles are right, so that angle is $\pi - A$. The three widths add up to $3\pi - \pi = 2\pi$, and adjacent sectors share a bounding ray. $\square$

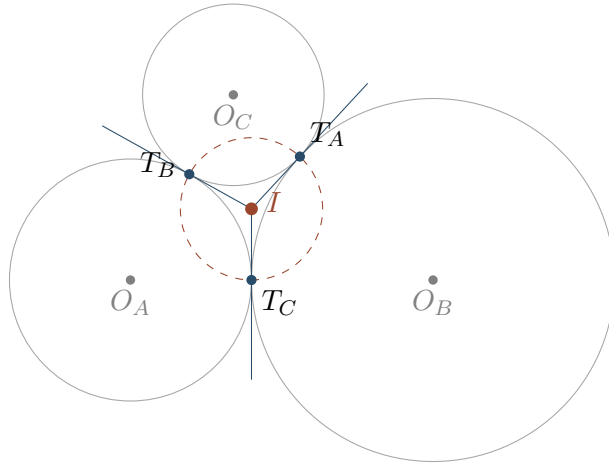


Figure 2: The incircle (dashed) passes through the three contact points. The rays from I through them are common tangents, and they cut the turn around I into the three sectors in which the three circles are seen.

So the directions of P, Q, R seen from I lie in three consecutive sectors (Figure 2). The triangle PQR contains I exactly when each of the three gaps between consecutive directions is less than π ; the gaps add up to 2π , so at most one of them can be larger. When the gap between P and Q is the large one, say that the side PQ misses I .

Lemma 2. *The events “ PQ misses I ”, “ QR misses I ” and “ RP misses I ” are disjoint, and their union is the event that PQR does not contain I . The side PQ misses I exactly when it crosses the common tangent IT_C beyond I , on the side of I away from T_C .*

Proof. The first sentence is the remark above. The circles about O_A and O_B lie on opposite sides of their common tangent, so PQ crosses it at one point Z . As a point runs from P to Q , its direction from I sweeps the angle $\angle PIQ < \pi$. If the gap from P to Q is less than π , the swept angle is that gap, which contains the ray IT_C , so Z lies on that ray. If the gap is larger than π , the swept angle is the rest of the turn, which contains the opposite ray, so Z lies beyond I . $\square$

So whether PQ misses depends only on where PQ crosses the tangent, and that is the subject of Theorem 2.

3 Where a random chord crosses the tangent

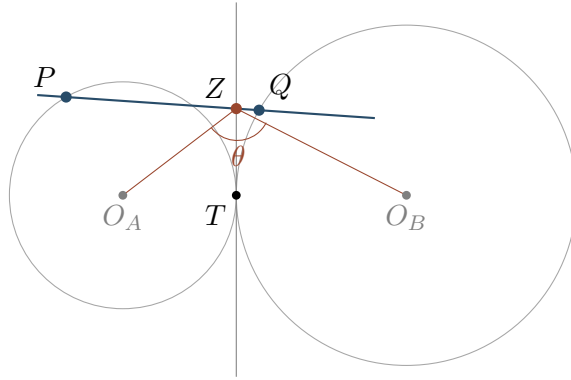


Figure 3: The chord PQ meets the common tangent at Z , from which the centres are seen at the angle θ . As Z climbs from T , θ falls from π towards $\frac{\pi}{2}$, uniformly in probability.

Put T at the origin, the tangent on the y -axis, $O_A = (-a, 0)$ and $O_B = (b, 0)$, and let $Z = (0, z)$. Write p and q for the angles that the segment TZ subtends at the two centres, so that

$$\tan p = \frac{z}{a}, \quad \tan q = \frac{z}{b}, \quad \theta = \pi - p - q. \quad (1)$$

Theorem 2 says that $s = p + q$ is uniform on $(-\frac{\pi}{2}, \frac{\pi}{2})$; equivalently, z has density

$$f(z) = \frac{1}{\pi} \left(\frac{a}{a^2 + z^2} + \frac{b}{b^2 + z^2} \right), \quad |z| < \sqrt{ab}, \quad (2)$$

which is the derivative of s/π . In words: the crossing point is spread like an even mixture of the two Cauchy laws centred at T with scales a and b , cut off where the chords stop reaching.

Proof of Theorem 2. We change variables from the pair of points to the line through them, and then count lines.

From points to lines. Describe the line PQ by its crossing height z and its direction $(\cos \omega, \sin \omega)$, with $|\omega| < \frac{\pi}{2}$. Moving (z, ω) by $(dz, d\omega)$ moves the point of the line at signed distance t from Z sideways by $\cos \omega dz + t d\omega$. Where the line cuts the first circle in a chord of half-length c_A , it meets the circle at an angle whose sine is c_A/a , so the angular position α_P of P on its circle changes by $(\cos \omega dz + t_P d\omega)/c_A$; similarly for Q . The Jacobian determinant is therefore

$$d\alpha_P d\alpha_Q = \frac{\cos \omega |PQ|}{c_A c_B} dz d\omega. \quad (3)$$

This is the classical density for pairs of points on two curves in integral geometry [6]. A line meeting both circles carries four pairs (P, Q) . Since Z lies between the two chords, the four

lengths $|PQ|$ add up to four times the distance between the chord midpoints, and that distance is the projection of $O_A O_B$ on the line, $(a+b)\cos\omega$. With (α_P, α_Q) uniform, of density $\frac{1}{4\pi^2}$, the four pairs together contribute

$$\frac{(a+b)\cos^2\omega}{\pi^2 c_A c_B} dz d\omega.$$

The chords. The tangent is the radical axis of the two circles, so Z has power z^2 with respect to each. The midpoint of the first chord is at distance $m_A = a\cos\omega + z\sin\omega$ from Z , so $c_A^2 = m_A^2 - z^2$, and with $z = a\tan p$ a short expansion gives

$$c_A^2 = \frac{a^2 \cos\omega \cos(\omega - 2p)}{\cos^2 p}, \quad c_B^2 = \frac{b^2 \cos\omega \cos(\omega + 2q)}{\cos^2 q}.$$

Now put $\omega = (p - q) + \eta$ and $s = p + q$. Then $\cos(\omega - 2p)\cos(\omega + 2q) = \cos(\eta - s)\cos(\eta + s) = \cos^2 s - \sin^2 \eta$, and the chords exist exactly when $|\eta| < \frac{\pi}{2} - |s|$. In particular lines through Z meet both circles only when $|s| < \frac{\pi}{2}$, that is $z^2 < ab$.

The substitution. For fixed z , put

$$u = \arcsin \frac{\sin \eta}{\cos s}, \quad du = \frac{\cos \eta d\eta}{\sqrt{\cos^2 s - \sin^2 \eta}}. \quad (4)$$

As η runs over $(-\frac{\pi}{2} + |s|, \frac{\pi}{2} - |s|)$, u runs over $(-\frac{\pi}{2}, \frac{\pi}{2})$. The probability element becomes

$$K(z) \frac{\cos\omega}{\cos\eta} dz du = K(z) (\cos(p-q) - \sin(p-q)\tan\eta) dz du, \quad K(z) = \frac{(a+b)\cos p \cos q}{\pi^2 ab}. \quad (5)$$

The second term is odd in η , hence in u , and integrates to zero; the first integrates over u to $\pi K(z) \cos(p-q)$. Finally, from $a = z \cot p$ and $b = z \cot q$,

$$\pi K(z) \cos(p-q) = \frac{\sin(p+q)\cos(p-q)}{\pi z} = \frac{\sin 2p + \sin 2q}{2\pi z} = \frac{1}{\pi} \left(\frac{\cos^2 p}{a} + \frac{\cos^2 q}{b} \right) = \frac{1}{\pi} \frac{ds}{dz},$$

which is (2). So s is uniform on $(-\frac{\pi}{2}, \frac{\pi}{2})$; reflecting in the line of centres gives the two signs of z equal probability, and $\theta = \pi - s$ is uniform on $(\frac{\pi}{2}, \pi)$ when $z > 0$. $\square$

The proof is a computation, and I do not know a way to see the uniformity directly. The substitution (4) is the one honest piece of luck: it makes the inner integral equal to π for every z .

4 Proof of Theorem 1

Apply Theorem 2 to the circles about O_A and O_B , measuring z along $T_C I$ towards I . The incentre is at height $\rho = T_C I$, the inradius, and by Lemma 2 the side PQ misses I exactly when $z > \rho$, that is when $\theta < \angle O_A I O_B$. The bisectors give $\angle O_A I O_B = \pi - \frac{A}{2} - \frac{B}{2} = \frac{\pi}{2} + \frac{C}{2}$. (This is also why $\rho < \sqrt{ab}$: the incentre lies within the range of crossing points.) Hence

$$\Pr(PQ \text{ misses } I) = \Pr(z > 0) \cdot \Pr\left(\theta < \frac{\pi}{2} + \frac{C}{2} \mid z > 0\right) = \frac{1}{2} \cdot \frac{C/2}{\pi/2} = \frac{C}{2\pi}.$$

In words, each side misses with probability equal to the opposite angle of the triangle of centres, measured in turns. By Lemma 2 the three misses are disjoint and make up the whole of “ I not in PQR ”, so

$$\Pr(I \notin PQR) = \frac{A+B+C}{2\pi} = \frac{1}{2}.$$

For three equal circles each side misses with probability $\frac{1}{6}$, which matches the integral in the equal case [1, 2].

Remark. There is a curious companion. The incircle meets each circle at right angles, and it cuts off from the circle about O_A an arc of angle A (the arc between T_B and T_C). So the vertex P lands inside the incircle with probability $\frac{A}{2\pi}$, and the side QR misses I with the same probability. Summing, $\Pr(I \notin PQR)$ equals the expected number of vertices inside the incircle. The two events are not the same event, and a direct correspondence between them would be the symmetry argument that Dan asked for.

5 A uniform point on the sphere

The proof threw away the coordinate u after integrating over it. Keeping it gives more.

Theorem 3. *With $s = p + q$ and u as in (4), put $v = |u|$, $L = 2s$ and $H = \frac{4v}{\pi} - 1$. Then the point of the unit sphere with longitude L and height H ,*

$$S(P, Q) = (\sqrt{1 - H^2} \cos L, \sqrt{1 - H^2} \sin L, H),$$

is uniformly distributed on the sphere.

Proof. In (5), the lines with directions η and $-\eta$ give the same v , and their elements add to $2K(z) \cos(p - q) dz dv = \frac{2}{\pi^2} ds dv$, with no η left over. So (s, v) is uniform on $(-\frac{\pi}{2}, \frac{\pi}{2}) \times (0, \frac{\pi}{2})$, and (L, H) is uniform on $(-\pi, \pi) \times (-1, 1)$. By Archimedes' theorem on the sphere and its circumscribed cylinder, the area element of the unit sphere in longitude and height is $dL dH$, so S is uniform. $\square$

Each pair (P, Q) of the map comes from one of eight pairs with the same (s, v) : two lines through Z and four pairs on each. The map does not preserve area sheet by sheet; only the eight sheets together do.

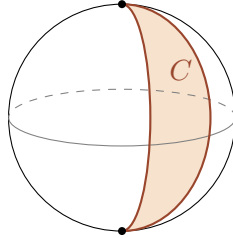


Figure 4: The event “ PQ misses I ” is the lune of longitudes $L > \pi - C$, of angle C , which covers the fraction $\frac{C}{2\pi}$ of the sphere.

On the sphere the miss events become lunes. The side PQ misses I when $z > \rho$, that is when $s > \frac{A}{2} + \frac{B}{2}$, or $L > \pi - C$: the lune between the meridians at $\pi - C$ and π , of angle C (Figure 4). Its area is $\frac{C}{2\pi}$ of the sphere. The three sides give lunes of angles A, B, C , one on each of three spheres, and as the angles of a triangle they add up to a hemisphere's worth.

6 What remains

The theorem is proved for all radii, but the question as asked is still open: a proof that shows the $\frac{1}{2}$ without integrating. Theorem 3 narrows the target. It would be enough to see, without computation, why the eight-sheeted map S preserves area; or why, in the remark of Section 4, a side misses as often as the opposite vertex falls inside the incircle.

Verification. The algebra and calculus of Sections 2 to 5 are formally verified in Lean 4 with Mathlib [7]: the chord identities, the integral of (4) over its whole range (equal to π), the density (2) and its distribution function, the half-angle identity behind $A + B + C = \pi$, the final sum, and the folding step of Theorem 3. The change of variables (3) as a statement about measures, and the plane geometry of Lemmas 1 and 2, are checked by hand and by simulation: an independent program confirms Theorem 1 for four radius triples, the per-side probabilities, the uniform angle, and the joint law of (s, v) . Code and proofs are at <https://github.com/jvvk/mathematics/tree/main/tangent-circles-incentre>.

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